K.BALAJI

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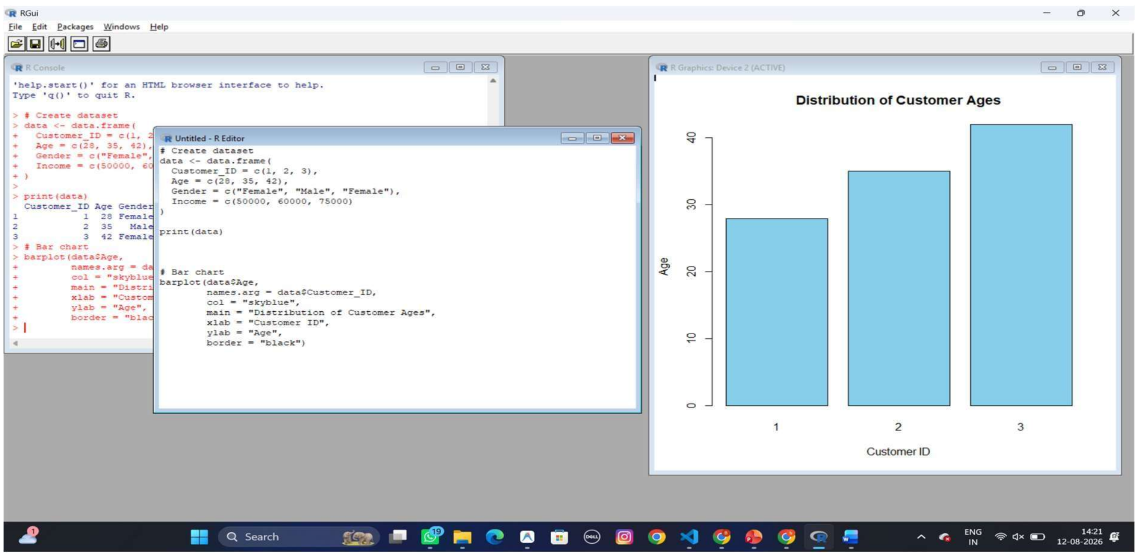
DATA HANDLING AND VISUVALIZATION LAB 16-20

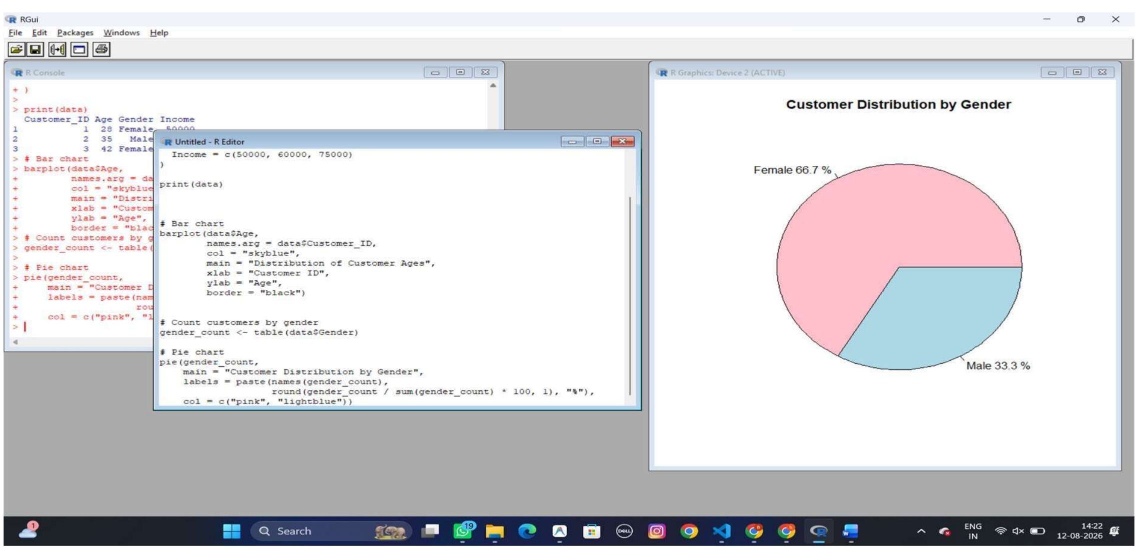
16) Customer Demographics Analysis

Dataset: Customer Demographics

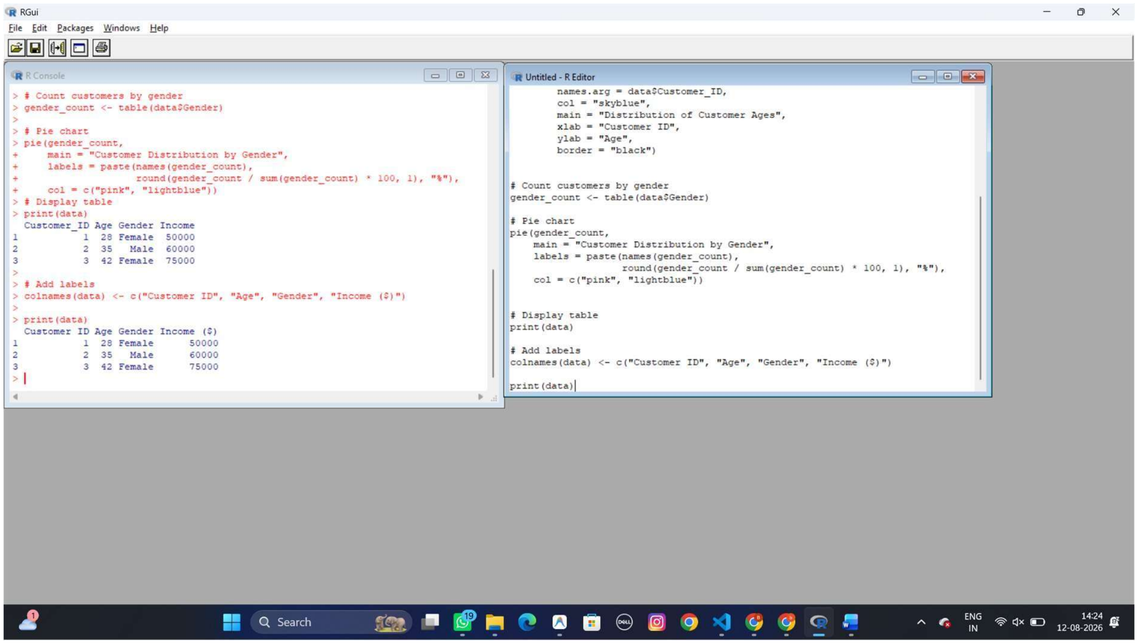
1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.



2. Generate a pie chart to represent the distribution of customers by gender.



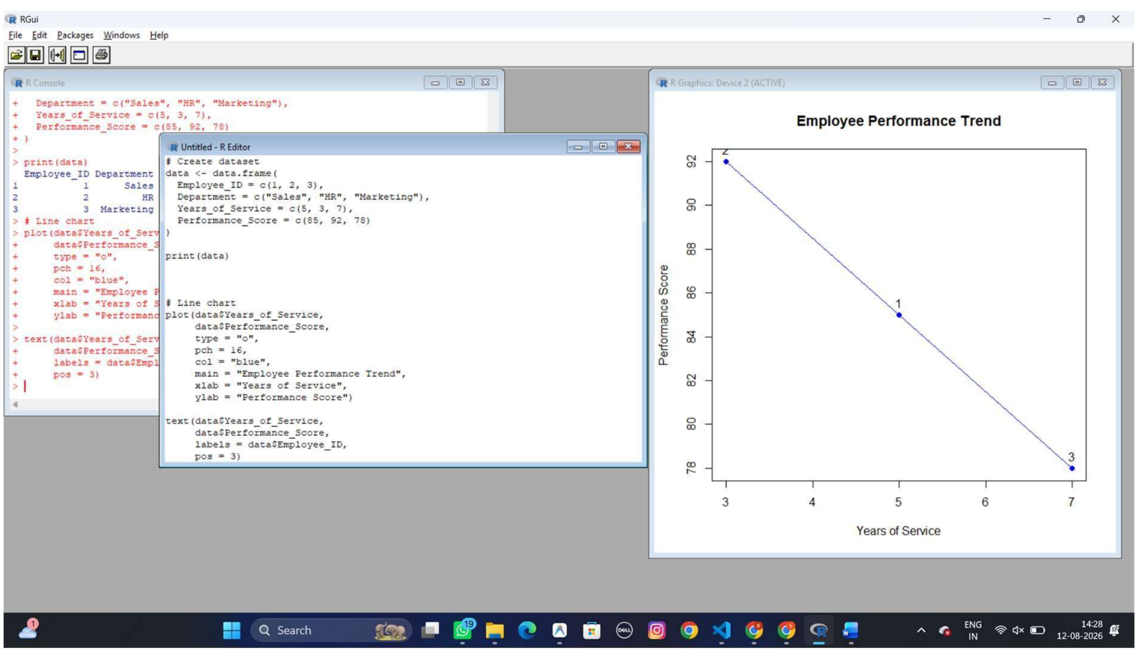
3. Build a table to show the demographic information for each customer. Label the table elements.



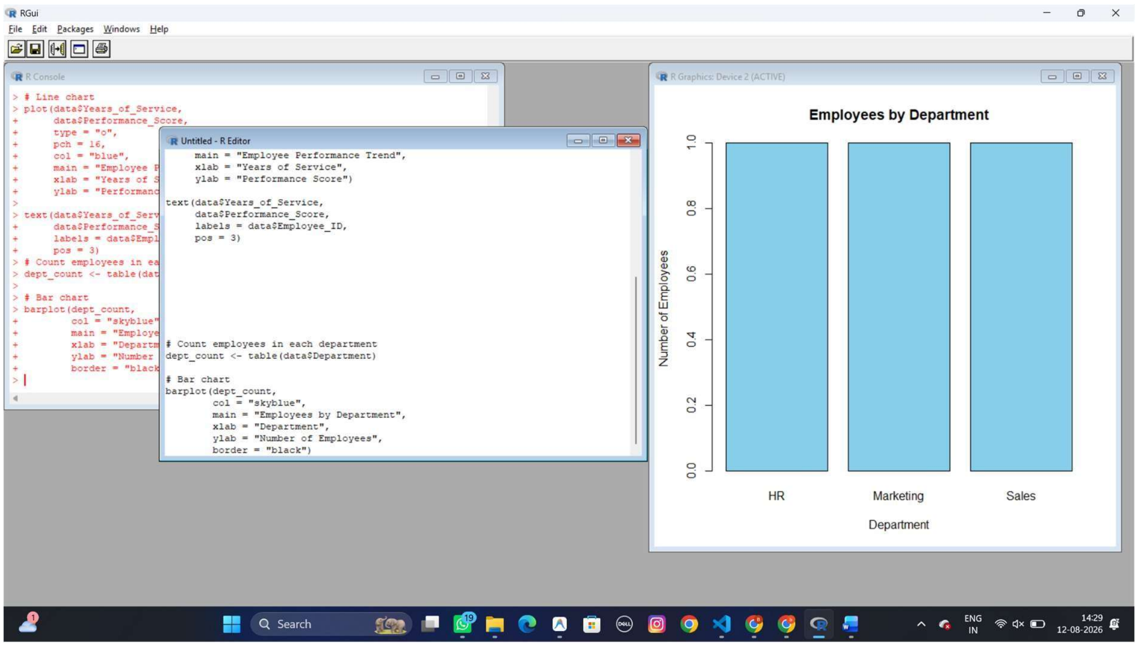
17: Employee Performance Analysis

Dataset: Employee Performance

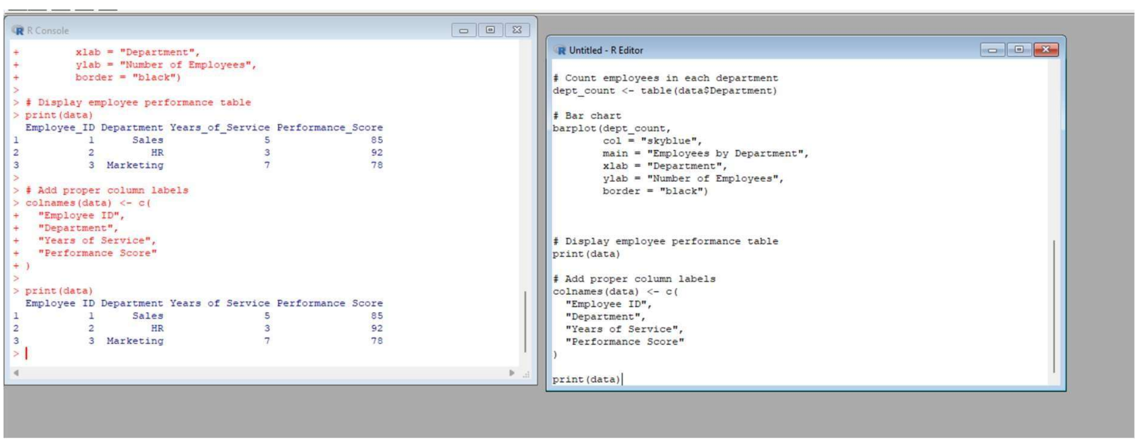
1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.



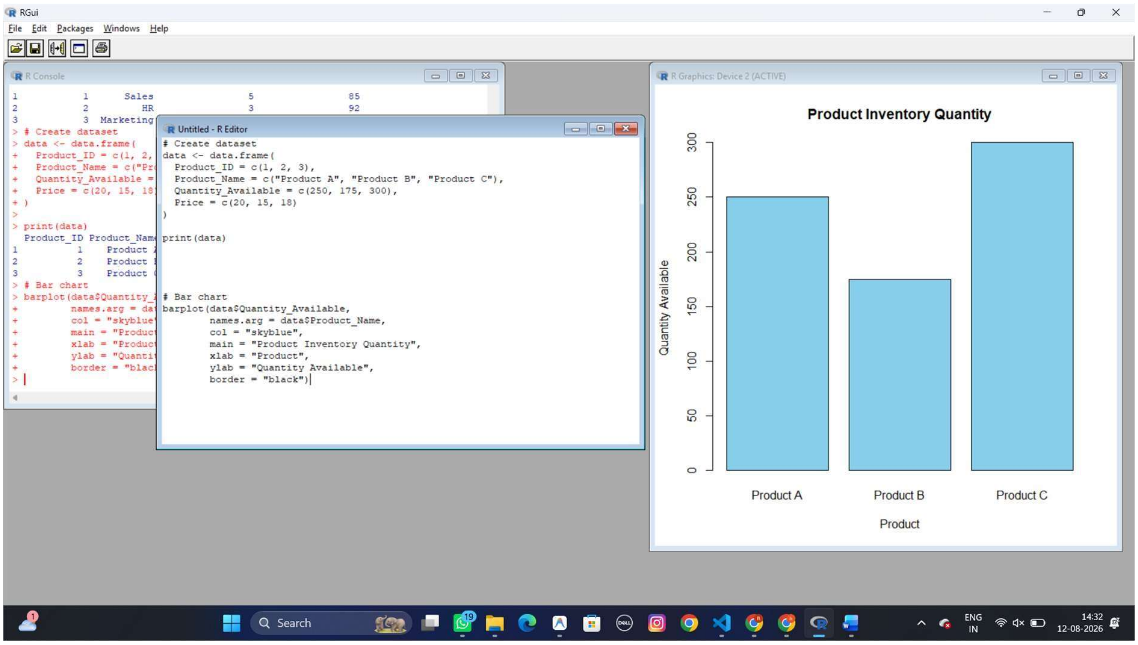
3. Build a table to display the performance data for each employee. Label the table elements.



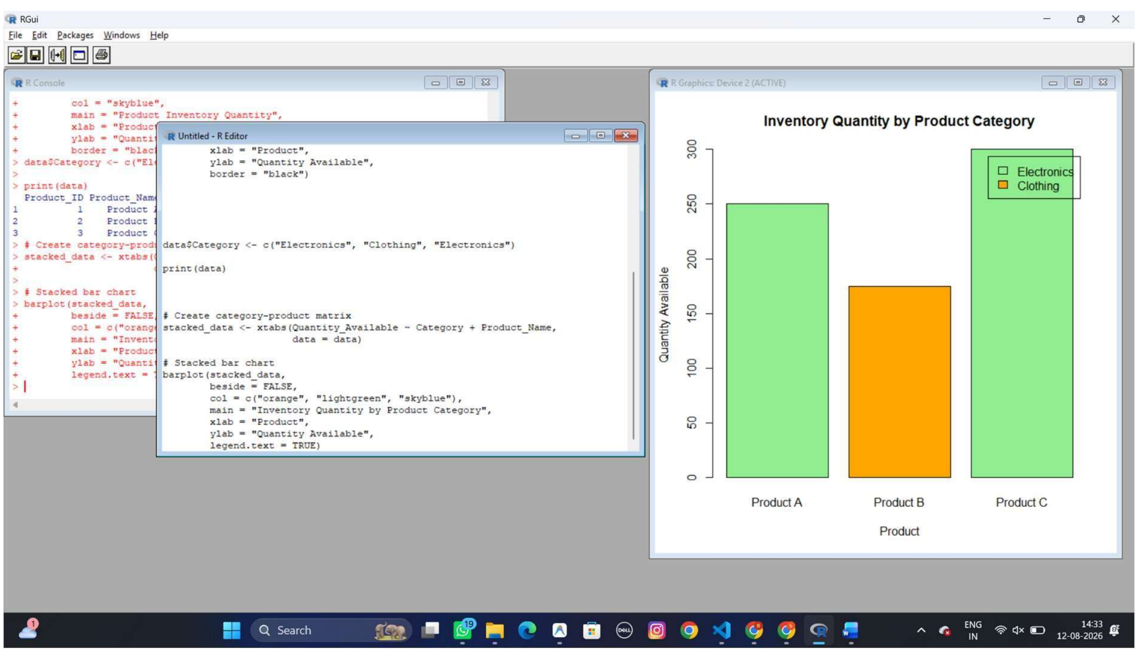
18: Product Inventory Management

Dataset: Product Inventory

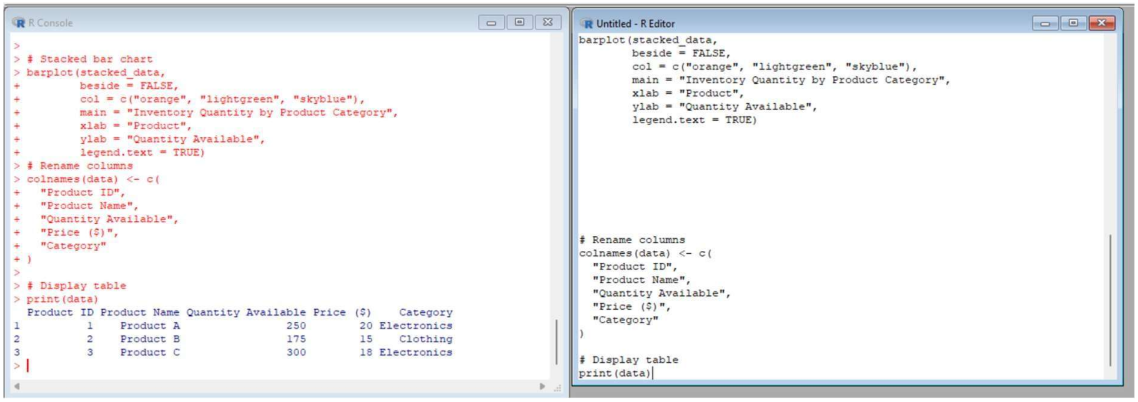
1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.



2. Generate a stacked bar chart to show the quantity of each product within different product categories.



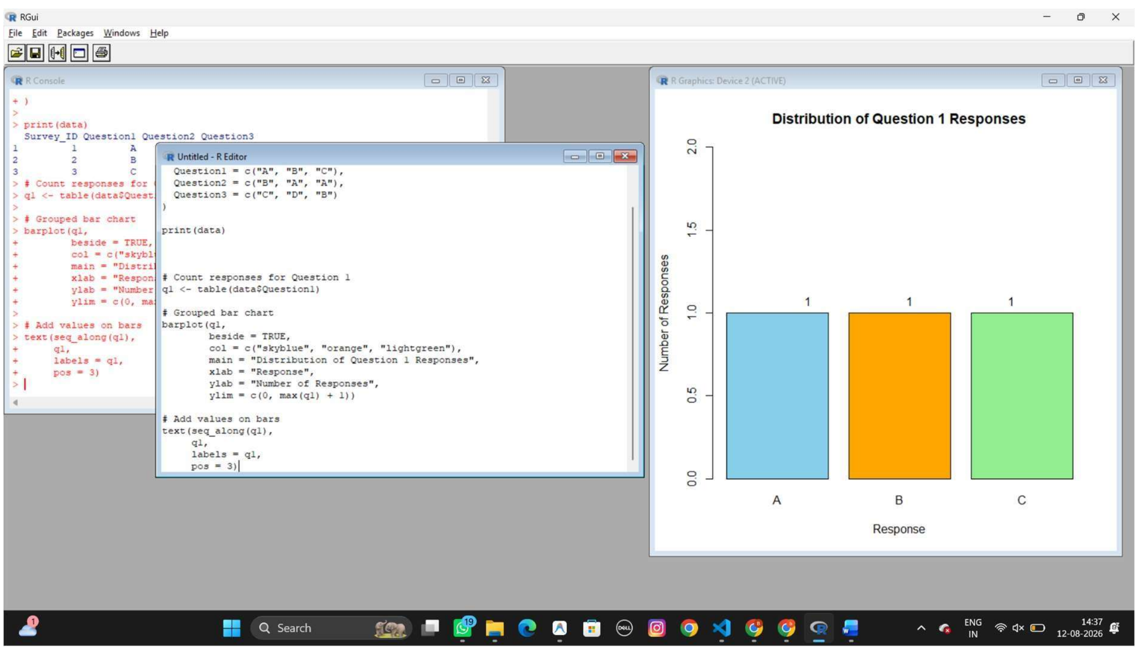
3. Build a table to show the inventory data for each product.
Label the table elements.



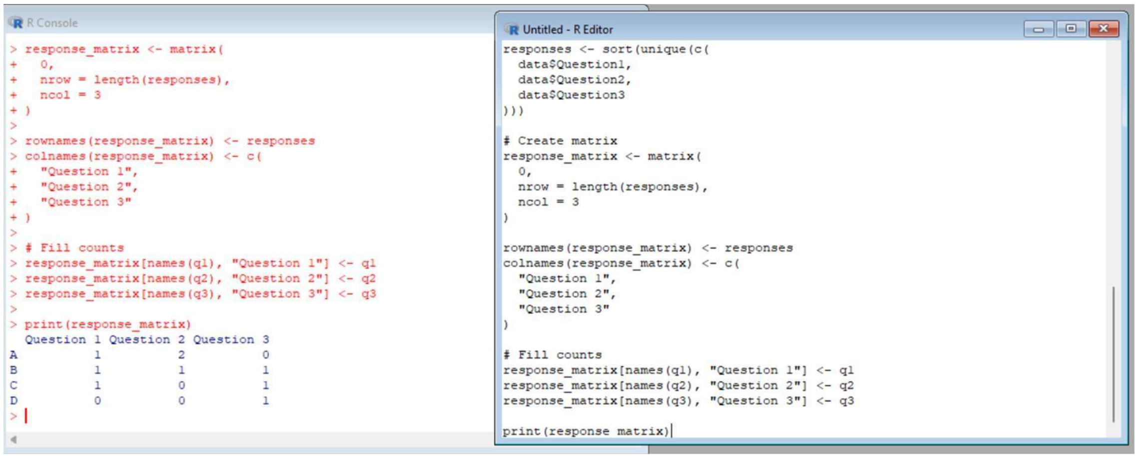
Set 19: Survey Responses Analysis

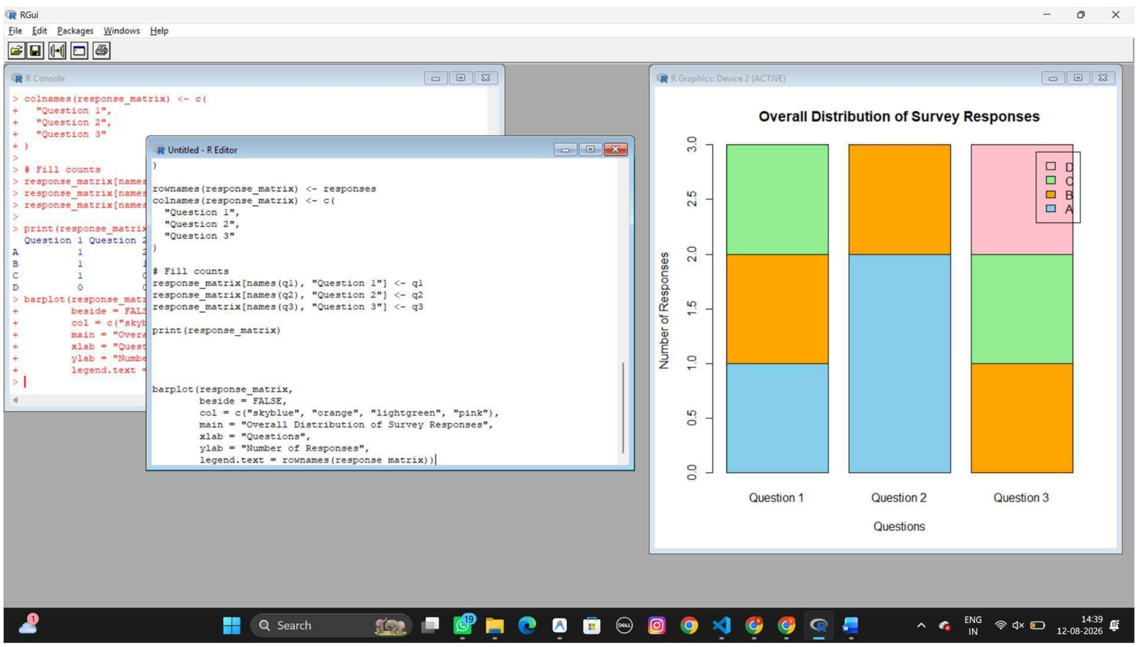
Dataset: Survey Responses

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

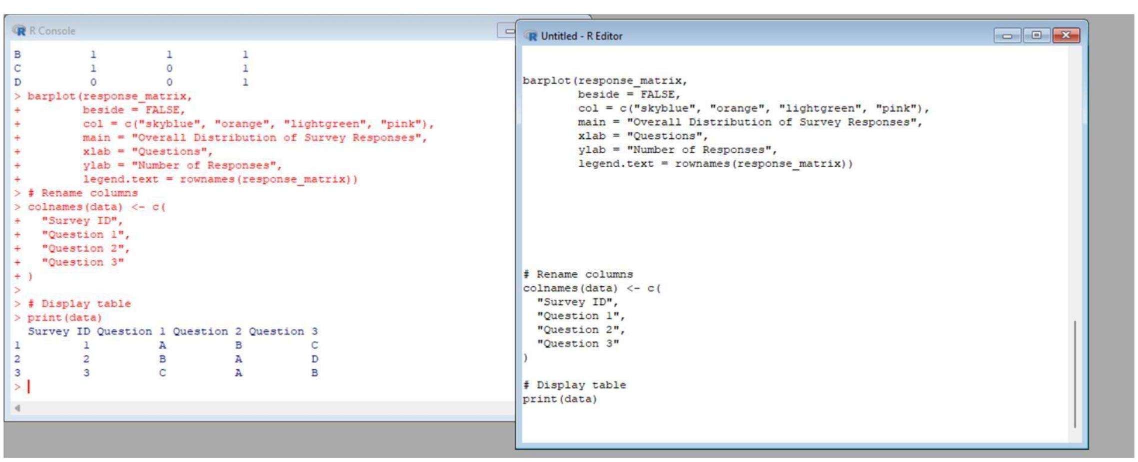


2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.





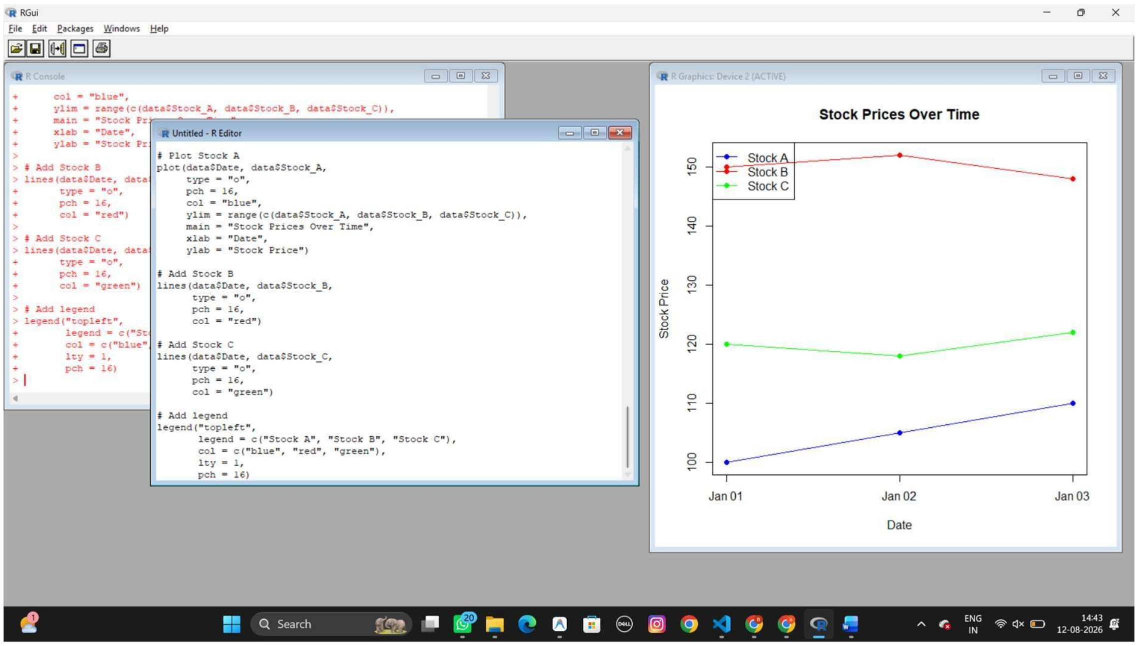
3. Build a table to show the survey response data for each survey. Label the table elements.



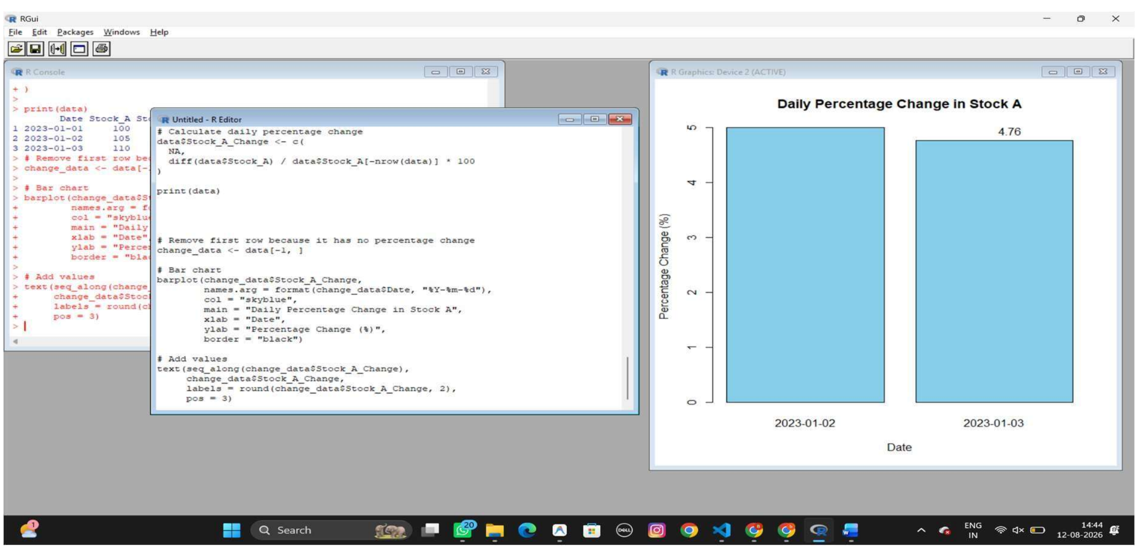
Set 20: Stock Analysis

Dataset: Stock Prices

1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.



2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.



3. Build a table to display the stock price data for each company over the given period. Label the table elements.

R Console

```
3 2023-01-03     110     148     122        4.761905
> # Remove first row because it has no percentage change
> change_data <- data[-1, ]
>
> # Bar chart
> barplot(change_data$Stock_A_Change,
+         names.arg = format(change_data$Date, "%Y-%m-%d"),
+         col = "skyblue",
+         main = "Daily Percentage Change in Stock A",
+         xlab = "Date",
+         ylab = "Percentage Change (%)",
+         border = "black")
>
> # Add values
> text(seq_along(change_data$Stock_A_Change),
+      change_data$Stock_A_Change,
+      labels = round(change_data$Stock_A_Change, 2),
+      pos = 3)
> # Display stock price table
> print(data)
        Date Stock_A Stock_B Stock_C Stock_A_Change
1 2023-01-01     100     150     120             NA
2 2023-01-02     105     152     118       5.000000
3 2023-01-03     110     148     122       4.761905
>
```

Untitled - R Editor

```
)

print(data)




# Remove first row because it has no percentage change
change_data <- data[-1, ]

# Bar chart
barplot(change_data$Stock_A_Change,
        names.arg = format(change_data$Date, "%Y-%m-%d"),
        col = "skyblue",
        main = "Daily Percentage Change in Stock A",
        xlab = "Date",
        ylab = "Percentage Change (%)",
        border = "black")

# Add values
text(seq_along(change_data$Stock_A_Change),
     change_data$Stock_A_Change,
     labels = round(change_data$Stock_A_Change, 2),
     pos = 3)


# Display stock price table
print(data)
```